

3. Users manually grant runtime permissions (e.g., camera, microphone) only when the feature is first used.

Applying the CIA Triad to mobile devices

The CIA triad – Confidentiality, Integrity, Availability – is a foundational security model. Here’s how to translate each principle into mobile-specific actions.



Confidentiality: Keeping data private

Confidentiality means ensuring only authorized parties can access sensitive information.

Step-by-Step: Enabling device encryption

1. Verify encryption status

- Android: Settings → Security → Encryption & Credentials.
- iOS: All modern devices are encrypted by default when you set a passcode.

2. Use encrypted messaging apps

- Install Signal or WhatsApp for end-to-end encrypted chats.

3. Lock sensitive apps

- On Android, enable “App Lock” in Settings to require biometrics for specific apps (e.g., banking).
- On iOS, use Screen Time → App Limits to restrict access.

Example: Janardhan enables full-device encryption on her Android phone (Settings → Security → Encrypt Phone), sets a strong alphanumeric PIN, and switches all messaging to Signal for private communication.

Integrity: Ensuring data is untampered

Integrity means preventing unauthorized modification of data, ensuring its accuracy and trustworthiness.

Step-by-Step: Verifying app and OS integrity

1. Stick to official app stores

- Download only from Google Play Store or Apple App Store.

2. Check app signatures

- On Android, enable “Verify apps” in Settings → Security to block sideloaded APKs that fail signature checks.

3. Enable OS update verification

- Both iOS and Android cryptographically verify updates before installation.

Example: Chaitali disables unknown sources on his Android (Settings → Security → Unknown sources OFF) to prevent unsigned APK installations, ensuring all apps originate from Google Play’s vetted ecosystem.

Availability: Ensuring reliable access

Availability guarantees that authorized users can access their mobile devices and data exactly when needed – even in adverse scenarios. On mobile devices, this encompasses not only uptime and performance but also resilient access to critical apps and information during emergencies or outages.

Step-by-Step: Building an availability-first mobile strategy

1. Implement redundant backups

■ Automated cloud backups:

- Android: Settings → System → Backup → Enable Google Drive backup (apps, call history, contacts, settings).
- iOS: Settings → [Your Name] → iCloud → iCloud Backup → Back Up Now; toggle on “iCloud Backup”.

■ Local encrypted backups (Optional):

- Android: Use adb backup or third-party tools (e.g., Titanium Backup) with device encryption enabled.
- iOS: Connect to a trusted PC/Mac → Finder or iTunes → Encrypt iPhone backup → Back Up Now.

2. Enable offline access for critical apps

■ Download offline maps & documents:

- Google Maps: Settings → Offline maps → Select Your Own Map → Download (ensures navigation without connectivity).
- Productivity Apps: In Dropbox/OneDrive apps, mark key folders “Available Offline” so files remain accessible without internet.

3. Maintain battery health and power redundancy

■ Battery optimization:

- Android: Settings → Battery → Battery Saver → Schedule or turn on at 15%.
- iOS: Settings → Battery → Low Power Mode → Toggle On.

■ Portable power solutions:

- Carry a trusted power bank (10,000 mAh+) and USB-C or Lightning cable.
- Use solar-powered chargers for extended field use.